SWIM(Title undefined for now)

Linhan Fan[1], Zhutao Wang[1], Hua He[1], Yafeng Niu[1*]

[1*]School of Mechanical Engineering, Southeast University, Nanjing, China.

*Corresponding author(s). E-mail(s): nyf@seu.edu.cn;
Contributing authors: linhanfan@seu.edu.cn; xxxx@seu.edu.cn;
xxxx@seu.edu.cn;

Abstract

1 Introduction

Citation Test [1, 2]

Background of the ultrasound mid-air haptic is presented in Supplementary Note 1.

2 Results

3 Discussion

4 Methods

Acknowledgments

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References

[1] Reardon, G., Dandu, B., Shao, Y. & Visell, Y. Shear shock waves mediate haptic holography via focused ultrasound. *Science Advances* (2023).

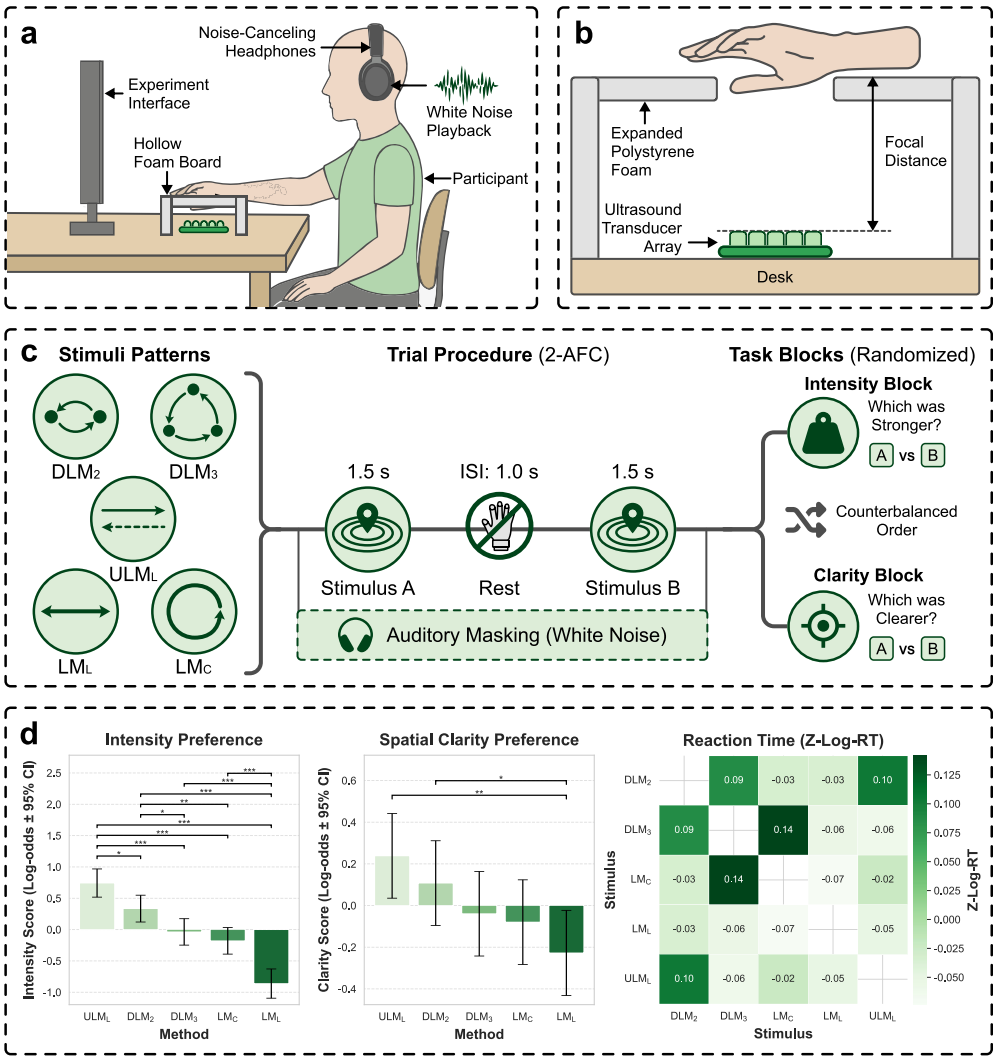

Fig. 1 Experimental design and psychophysical evaluation of spatiotemporal modulation patterns. a, Schematic of the experimental setup. Participants were seated with their hand positioned on a hollow foam board over the ultrasound phased array, wearing active noise-canceling headphones playing white noise to eliminate acoustic cues. **b**, Detailed view of the hand-device interface showing the acoustic window and focal alignment. **c**, Illustration of the five modulation patterns (DLM_2, DLM_3, ULM_L, LM_L, LM_C) and the two-alternative forced choice (2-AFC) protocol. The experiment consisted of two counterbalanced blocks evaluating intensity perception and spatial clarity. **d**, Psychophysical results ($n = 26$). Left and Center: Relative preference scores (log-odds) for intensity and spatial clarity derived from a Bayesian mixed-effects Bradley-Terry model. Error bars represent 95% confidence intervals. Statistical significance is indicated by asterisks (*$P < 0.05$, ***$P < 0.001$; Wald test). Right: Matrix of standardized logarithmic reaction times (Z-Log-RT) for pairwise comparisons, where darker colors indicate longer decision times and higher discrimination difficulty.

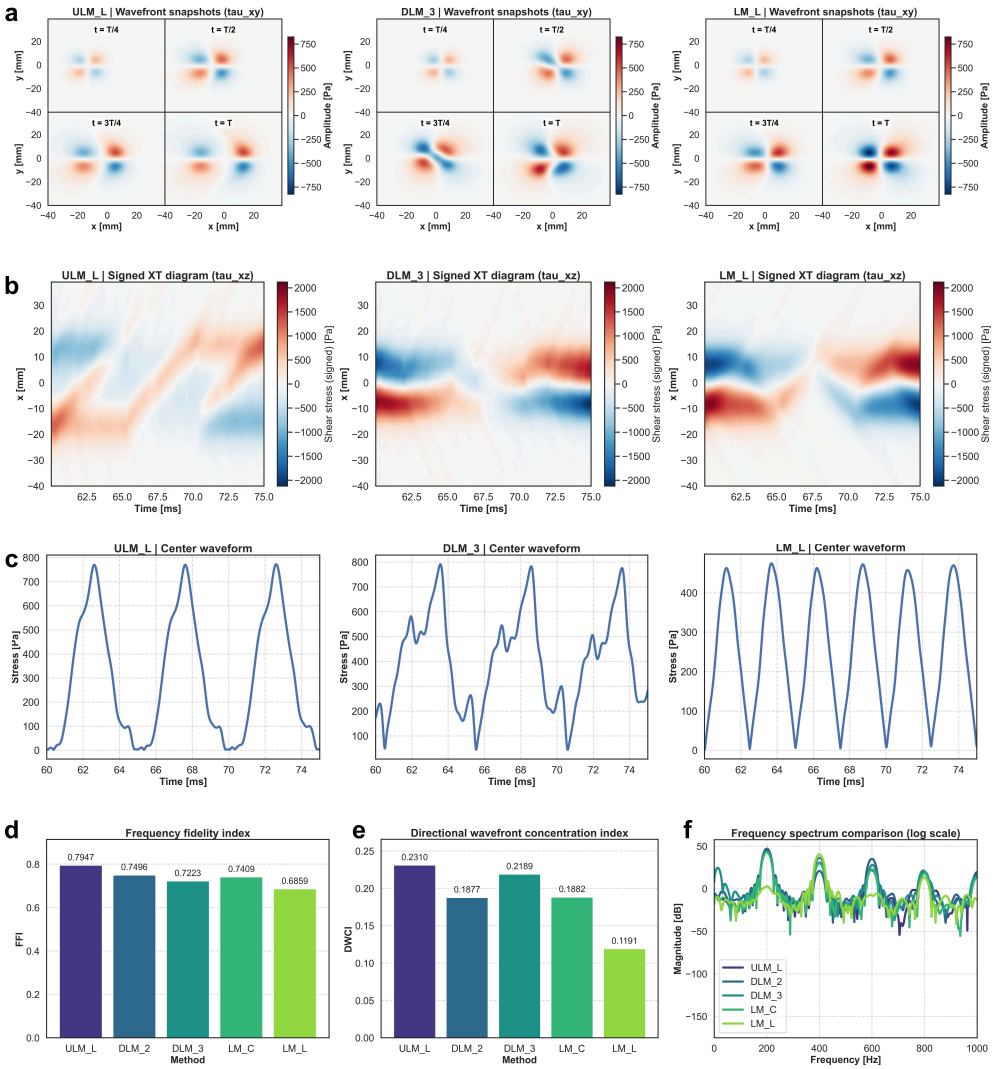

Fig. 2 Elastodynamic simulation of cutaneous shear wave propagation and stress distribution. a, Spatial distributions of peak shear stress in the xy-plane for representative modulation patterns (ULM_L, DLM_3, and LM_L), illustrating distinct energy concentration profiles. **b**, Spatiotemporal (x-t) diagrams of the signed shear stress along the modulation trajectory ($y = 0$), visualizing the constructive interference and wave propagation dynamics. **c**, Temporal waveforms of shear stress at the focal center during the steady-state phase (last 15 ms). **d**, Normalized lateral focusing profiles comparing the spatial resolution across all modulation strategies. The full width at half maximum (FWHM) is annotated for each pattern. **e**, Cross-sectional distribution of the root-mean-square (RMS) shear stress along the lateral axis ($y = 0$). **f**, Spectral analysis of the induced shear waves (log scale), highlighting the fundamental frequency (200 Hz) and harmonic components generated by different spatiotemporal dynamics.

[2] Takahashi, R., Hasegawa, K. & Shinoda, H. Tactile stimulation by repetitive lateral movement of midair ultrasound focus. *IEEE Transactions on Haptics* (2020).